Introduction

Splunk is a Security Information and Event Management (SIEM) platform used for collecting, indexing, monitoring, and analyzing machine-generated data in real time.

This project demonstrates hands-on experience with Splunk by building dashboards for SSH logs, web traffic logs, and Cloudflare logs. The objective of these projects was to understand log ingestion, data parsing, SPL (Search Processing Language), dashboard creation, and security monitoring techniques commonly used in SOC environments.

Splunk Architecture Overview

Splunk architecture mainly consists of three core components

Forwarder: A Forwarder collects logs from endpoints or servers and sends them to the Splunk Indexer.

Types of Forwarders

Universal Forwarder (Lightweight): A lightweight Splunk agent that collects and forwards logs with minimal resource usage.

Heavy Forwarder (Heavyweight): A full Splunk instance capable of parsing, filtering, transforming, and routing logs before forwarding them to indexers.

Indexer

The Indexer receives logs from forwarders, processes the data, and stores it in indexes for searching and analysis.

Search Head

The Search Head provides the user interface where analysts can search logs, create dashboards, generate alerts, and perform investigations using SPL queries.

Splunk Deployment Models

Splunk can be deployed using different architectures depending on the size, scalability, and requirements of the environment.

1. Single Instance Deployment

A Single Instance deployment contains all Splunk components on one server.

Components Included

- Forwarder
- Indexer
- Search Head

Features

- Simple setup
- Easy to manage
- Suitable for learning and small environments
- Limited scalability

Use Cases

- Personal labs
- Testing environments
- Small organizations

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2. Distributed Deployment

In a Distributed Deployment, Splunk components are separated across multiple servers.

Architecture

- Forwarders collect logs
- Indexers store and process data
- Search Heads handle searches and dashboards

Features

- Better performance
- Scalable architecture
- Supports larger environments
- Improved resource management

Use Cases

- Medium to large organizations
- SOC environments
- Enterprise monitoring systems

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3. Clustered Deployment

A Clustered Deployment uses multiple indexers and search heads to provide high availability and redundancy.

Types of Clustering

Indexer Clustering

Multiple indexers replicate data for fault tolerance and data availability.

Search Head Clustering

Multiple search heads provide load balancing and high availability for users.

Features

- High availability
- Data redundancy
- Load balancing
- Fault tolerance
- Enterprise-grade scalability

Use Cases

- Large enterprises
- Critical production environments
- High-volume security monitoring

Simple Comparison

Deployment Model	Complexity	Scalability	High Availability	Best For
Single Instance	Low	Low	No	Learning & Small Labs
Distributed	Medium	High	Partial	Medium/Large Organizations
Clustered	High	Very High	Yes	Enterprise Environments